ModelIt!

Teacher's Guide

Where Did the Puddle Go?

How Heating and Cooling Change Matter

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-PS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-PS1-4

Construct an argument with evidence that some changes caused by heating or cooling can be reversed and some cannot.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students construct an explanation for where puddle water goes, using evidence from their model about heating and evaporation.
Disciplinary Core Idea	**PS1.B Chemical Reactions** — Heating or cooling a substance may cause changes that can be observed. Sometimes these changes are reversible, and sometimes they are not.
Crosscutting Concept	**Cause and Effect** — Students identify that heat from the sun causes water to evaporate, and cooling causes water vapor to condense.

CAST Testing Alignment

MULTIPLE CHOICE:

A puddle on the playground gets smaller and smaller on a sunny day. Where did the water go?

CONSTRUCTED RESPONSE:

Explain what happens to water in a puddle when the sun heats it up. Use the words 'evaporation' and 'water vapor' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water is always on the move! It travels from puddles to the sky and back again in a never-ending journey. Understanding how heating and cooling change water helps us explain weather, clouds, and rain.

Evaporation: Liquid to Gas

When the sun heats water, it gives the water energy. The water molecules at the surface start moving faster and faster until they escape into the air as water vapor — an invisible gas. This process is called evaporation. It happens all the time, but it happens faster when it is hot, windy, or dry.

Condensation: Gas to Liquid

When water vapor rises into the sky, it cools down. As it cools, the water molecules slow down and clump back together into tiny liquid droplets. This is condensation. You can see condensation on a cold glass of water on a hot day — those water drops on the outside come from water vapor in the air cooling down on the cold glass.

Reversible Changes

Evaporation and condensation are reversible changes. The water can go from liquid to gas and back to liquid over and over again. Freezing and melting are also reversible — ice melts into water, and water can freeze back into ice. Not all changes are reversible though — you cannot un-bake a cookie!

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System

Students identify sun heat as an external component and evaporation speed and puddle size as internal components that change when the sun heats the puddle.

E — Establish Relationships

Students determine that more sun heat causes faster evaporation (positive), and faster evaporation causes the puddle to get smaller (negative — puddle size decreases).

V — Visualize & Model

Students build a model showing how the sun's heat drives evaporation, which shrinks the puddle over time.

E — Evaluate Outcomes

Students run sunny day and cloudy day scenarios to compare how fast puddles disappear under different heat conditions.

R — Revise & Extend

Students add wind speed or puddle depth to explore other factors that affect how quickly puddles evaporate.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever seen a puddle after it rains? What happens to it the next day?"*

Watch For

Let students share their observations about puddles. Show a time-lapse of a puddle evaporating if available.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to solve a mystery — where do puddles go when they dry up?"*

Watch For

Read objectives together. Introduce evaporation, water vapor, condensation, and reversible with gestures.

Slide 3: The Big Question

3 min

Say: *"When a puddle disappears on a sunny day, does the water just vanish? Where does it go?"*

Watch For

Think-pair-share: Where do you think puddle water goes? Collect ideas on the board.

Slide 4: Let's Build a Model!

3 min

Say: *"Let us build a model to track where the puddle water goes — step by step."*

Watch For

Preview each LEVER step. Remind students that scientists use models to explain things they cannot easily see.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What makes a puddle disappear? What parts are involved? Which part do we control?"*

Watch For

Guide students to sort Sun Heat as external and Evaporation Speed and Puddle Size as internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When there is more sun heat, does the puddle evaporate faster or slower? Does the puddle get bigger or smaller?"*

Watch For

Draw positive arrow from Sun Heat to Evaporation Speed. Draw negative arrow from Evaporation Speed to Puddle Size.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test our model! First a super sunny day, then a cloudy day. What do you predict?"*

Watch For

Students predict then run scenarios. Compare sunny vs. cloudy results. Discuss why the difference.

Slide 8: What Did We Discover?

7 min

Say: *"The puddle water did not disappear — it just changed into something invisible! Who can tell me what it became?"*

Watch For

Breathe on a cold window or mirror to show condensation. Connect evaporation and condensation as reverse processes.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sun Heat
- **INTERNAL:** Evaporation Speed, Puddle Size

Relationships

- **Sun Heat to Evaporation Speed:** POSITIVE (+) — More sun heat gives the water more energy, making it evaporate faster. On a hot day the puddle dries up quickly because the sun is heating the water and turning it to vapor faster.
- **Evaporation Speed to Puddle Size:** NEGATIVE (-) — When evaporation is faster, the puddle gets smaller more quickly. The water is leaving the puddle and going into the air, so the puddle shrinks.

Key Discoveries

- Heat from the sun causes puddle water to evaporate — it turns into invisible water vapor
- On hot sunny days, puddles disappear faster because evaporation speed increases
- The water did not disappear forever — it went into the air as water vapor
- When water vapor cools down, it can turn back into liquid water (condensation) — this change is reversible

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The puddle water is gone forever"	The water did not disappear — it changed form! It turned into an invisible gas called water vapor through evaporation. The water is still in the air around us, and it can turn back into liquid when it cools down.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Only boiling water makes steam or vapor"	Water does not have to boil to evaporate! Evaporation happens at any temperature — even on cool days. The sun slowly gives water enough energy for some molecules to escape into the air. Boiling just makes it happen much faster.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Water vapor is the same as clouds or steam you can see"	Real water vapor is completely invisible — you cannot see it at all. The white mist you see above a boiling pot or in clouds is actually tiny liquid water droplets, not vapor. True water vapor is a gas hiding in the air.